# MANTA

*One continuous ribbon. One gesture.*

*Cut like a baguette so it can be printed at home and still hold an adult.*



## IDEA

MANTA is a single thick ribbon: floor, dished seat, cantilevered backrest, forward-curled crown, tail-brace, back to the floor — no separate legs, no frame, no visible assembly. The name is literal: forked feet read as fins, the tail-brace as a fluke. The ocean makes most of the oxygen we breathe, and the manta ray is one of its health indicators — so the surface carries the story too: flutes that follow the ribbon's own motion, bold at the floor and easing toward the crown, capped where you sit.

## BIOMIMICRY AS STRUCTURE, NOT ORNAMENT

- **Wolff's law** — thickness follows the bending moment: 84 mm at the lumbar knot, 50 mm at the knee, a 32 mm blade at the crown.
- **Murray's law** — the lumbar, the one true junction, is the cube-root-sum of seat, backrest and tail; every transition filleted.
- **Spiral fibre** — the flutes follow the stress path and stiffen by corrugation; the section twists up to 20° through the backrest.
- **Shell curvature** — the closed section and dished seat carry load as a shell.
- **Fiddlehead scroll** — the crown curls forward and returns *through* the backrest: a closed loop.
- **Buttress roots** — each foot forks into two splayed prongs.

## PRINTING

41 segments (each ≤ 212 × 220 × 250 mm), auto-oriented for the least support: **almost support-free**, under 1 % of the filament (PrusaSlicer), none inside the holes — each is shaped for the way its part lies. PETG, 0.2 mm layers, 5 perimeters, 35 % gyroid. ~16.5 kg, ~460 h on one printer, ~90 €.

## ASSEMBLY — COMPUTED JOINTS, NO SCREWS

Cuts and joints are computed, not drawn by hand. The script slices the ribbon **crosswise** by length and curvature, keeping every cut ≥ 70 mm from the knee and lumbar, where bending peaks. Every joint is a full solid cross-section (7,000–17,000 mm²), epoxied, with a self-centring conical peg sized to the local thickness and a 2 mm socket wall. Only the two most-loaded lumbar joints add a solid Ø11 pin. At the crossing the returning band is trimmed to the backrest's shape and epoxied on as a saddle. Every part's number is engraved on a glue face. One command (manta_verify.py) re-checks it all: strength, a **virtual assembly** (pegs seated, 2 mm walls, no overlaps), bed fit, real-slicer support.

## STABILITY AND LOAD

First-order check (120 kg × 1.8 dynamic × 2.0 safety): every member passes; glued full-section joint 26×, weakest glued check 2.1×; with no glue at all, peg and pin still carry 0.9×. Tipping 0.56 sideways, 0.55–0.77 rearward; safe static load ~250 kg. Seat 429 × 364 mm, 428 mm high; 8/8 ergonomics checks pass.

| ~250 kg | 26× | ~16.5 kg | ~90 € |
|---|---|---|---|
| safe static load | glued full-section joint | finished weight | filament cost |

## OPEN DESIGN — NOTHING REPEATS

Nothing in nature repeats, so no two parts here do either: all 41 segments are geometrically unique. Change the tables and the script regenerates a leaner chair with its joints and checks; a 70 kg, half-filament variant is in the repo (link below).